

INTEGRATED PLATFORM FOR CROWDSOURCED OCEAN HAZARD REPORTING AND SOCIAL MEDIA ANALYTICS

A Report

*Submitted in partial fulfilment of
the Requirements for the
completion of*

THEME BASED PROJECT

**BACHELOR OF ENGINEERING
IN
INFORMATION TECHNOLOGY**

By

B. POORNIMA	1602-23-737-032
K. SAI REVANTH	1602-23-737-042
VAIBHAV SHAHI	1602-23-737-060

Under the guidance of
Dr. B KEZIA RANI
Associate Professor



Department of Information Technology

Vasavi College of Engineering (Autonomous)

ACCREDITED BY NAAC WITH 'A++' GRADE

(Affiliated to Osmania University and Approved by AICTE)

Ibrahim Bagh, Hyderabad-31

2026

Vasavi College of Engineering (Autonomous)

ACCREDITED BY NAAC WITH 'A++' GRADE

(Affiliated to Osmania University and Approved by AICTE)

Ibrahim Bagh, Hyderabad-31

Department of Information Technology



DECLARATION BY CANDIDATES

We, **B. Poornima, K. Sai Revanth, Vaibhav Shahi**, bearing hall ticket numbers **1602-23-737-032, 1602-23-737-042, 1602-23-737-060**, hereby declare that the project report entitled **"Integrated Platform for Crowdsourced Ocean Hazard Reporting and Social Media Analytics"** under the guidance of Dr. Burugula Kezia Rani, Associate Professor, Department of Information Technology, Vasavi College of Engineering, Hyderabad, is submitted in partial fulfilment of the requirement for the completion of Theme-based project, VI semester, Bachelor of Engineering in Information Technology.

This is a record of bonafide work carried out by us and the results embodied in this project report have not been submitted to any other institutes.

B. Poornima 1602-23-737-032
K. Sai Revanth 1602-23-737-042
Vaibhav Shahi 1602-23-737-060

Vasavi College of Engineering (Autonomous)

ACCREDITED BY NAAC WITH 'A++' GRADE

(Affiliated to Osmania University and Approved by AICTE)

Ibrahim Bagh, Hyderabad-31

Department of Information Technology



BONAFIDE CERTIFICATE

This is to certify that the project entitled "**Integrated Platform for Crowdsourced Ocean Hazard Reporting and Social Media Analytics**" being submitted by **B. Poornima, K. Sai Revanth, Vaibhav Shahi** bearing **1602-23-737-032, 1602-23-737-042, 1602-23-737-060**, in partial fulfilment of the requirements for the completion of Theme-based project of Bachelor of Engineering in Information Technology is a record of bonafide work carried out by them under my guidance.

Dr. Burugula Kezia Rani

Associate Professor

External Examiner

Dr. K. Ram Mohan Rao

Professor, HOD IT

ACKNOWLEDGEMENT

The satisfaction that accompanies the successful completion of the project would not have been possible without the kind support and help of many individuals. We would like to extend our sincere thanks to all of them.

It is with immense pleasure that we would like to take the opportunity to express our humble gratitude to **Dr. Burugula Kezia Rani, Associate Professor, Information Technology** under whom we executed this project. Their constant guidance and willingness to share their vast knowledge helped us understand this project in great depth and complete the assigned tasks successfully.

We are very much thankful to **Dr. K. Ram Mohan Rao, Professor and HOD, Information Technology**, for his kind support and for providing the necessary facilities to carry out the work.

We wish to convey our special thanks to **Dr. S. V. Ramana, Principal, Vasavi College of Engineering**, for his encouragement and for providing us with the necessary resources. Not to forget, we thank all faculty, non-teaching staff, and friends who directly or indirectly helped and supported us in completing this project in time.

We also express our sincere thanks to the Management for providing excellent facilities. Finally, we wish to convey our gratitude to our families who fostered all the support we needed throughout this project.

ABSTRACT

Coastal and ocean hazard events such as high waves, storm surges, and coastal flooding pose significant threats to lives and infrastructure in coastal communities. While national agencies like the Indian National Centre for Ocean Information Services (INCOIS) provide scientific early warning systems, there remains a critical gap in the integration of real-time citizen observations and unstructured social media data into a unified disaster response platform.

This project proposes an Integrated Platform for Crowdsourced Ocean Hazard Reporting and Social Media Analytics that bridges this gap by combining mobile-based crowdsourced hazard reporting, geospatial visualization, AI-driven social media monitoring, and an automated alert notification system into a single, cohesive platform accessible through mobile and web interfaces.

Citizens can submit geotagged hazard reports including images and videos via a Flutter-based mobile application, even in offline conditions, with automatic synchronization upon network restoration. Reports are stored in a PostgreSQL database with the PostGIS geospatial extension and visualized on interactive maps with dynamic hotspot detection on a React-based web dashboard for use by government officials and analysts.

The platform concurrently monitors social media platforms (Twitter and YouTube) and employs Natural Language Processing (NLP) techniques including sentiment analysis, keyword extraction, and hazard classification to identify and surface emerging hazard indicators from public discourse. Insights are displayed through an analytics dashboard that tracks trends, keyword frequencies, and public sentiment in near real time.

An automated alert system generates multi-level notifications (low, medium, high) when report density or social media signals exceed predefined thresholds, ensuring timely escalation to disaster management authorities. The system is designed for high availability (99% uptime), supports 500-1000 concurrent users, and ensures data security through HTTPS encryption, JWT authentication, and role-based access control.

TABLE OF CONTENTS

1. INTRODUCTION	10
1.1 Overview	10
1.2 Problem Statement	11
1.3 Motivation of Theme & Title.....	12
2. LITERATURE SURVEY	13
3. EXISTING SYSTEM	15
4. PROPOSED SOLUTION	17
4.1 System Design.....	17
4.1.1 Architecture Overview	17
4.1.2 UML Diagrams and their Descriptions.....	18
4.2 Functional Modules	36
4.2.1 Screenshots & Pseudocode	37
5. EXPERIMENTAL SETUP & IMPLEMENTATION.....	39
5.1 System Specifications	39
5.1.1 Hardware Requirements.....	39
5.1.2 Software Requirements	40
5.2 Datasets	40
5.3 Methodology / Algorithm	41
6. RESULTS	42
7. CONCLUSION & FUTURE SCOPE.....	43
8. REFERENCES.....	45

LIST OF FIGURES

<i>Fig No.</i>	<i>Figure Title</i>
<i>Fig 1</i>	<i>System Architecture</i>
<i>Fig 2</i>	<i>Use Case Diagram</i>
<i>Fig 3</i>	<i>Use Case Descriptions</i>
<i>Fig 4</i>	<i>Class Diagram</i>
<i>Fig 5</i>	<i>Sequence Diagram – Submit Hazard Report</i>
<i>Fig 6</i>	<i>Sequence Diagram – Social Media Monitoring & NLP Analysis</i>
<i>Fig 7</i>	<i>Sequence Diagram – Offline Report Synchronization</i>
<i>Fig 8</i>	<i>Collaboration Diagram – Submit Hazard Report</i>
<i>Fig 9</i>	<i>Collaboration Diagram – Social Media Monitoring & NLP Analysis</i>
<i>Fig 10</i>	<i>Collaboration Diagram – Offline Report Synchronization</i>
<i>Fig 11</i>	<i>Activity Diagram – Hazard Report Submission</i>
<i>Fig 12</i>	<i>Activity Diagram – Social Media Analytics Workflow</i>
<i>Fig 13</i>	<i>State Chart Diagram – Hazard Report Lifecycle</i>
<i>Fig 14</i>	<i>State Chart Diagram – Alert System Flow</i>
<i>Fig 15</i>	<i>Component Diagram</i>
<i>Fig 16</i>	<i>Deployment Diagram</i>
<i>Fig 17</i>	<i>Landing Page</i>
<i>Fig 18</i>	<i>User Dashboard</i>
<i>Fig 19</i>	<i>Community Dashboard</i>
<i>Fig 20</i>	<i>Triage Queue</i>
<i>Fig 21</i>	<i>Internal Communication Dashboard</i>

LIST OF TABLES

<i>Table No.</i>	<i>Table Title</i>
<i>Table 1</i>	<i>List of Abbreviations</i>
<i>Table 2</i>	<i>Comparison of Existing Systems</i>
<i>Table 3</i>	<i>Use Case Descriptions</i>
<i>Table 4</i>	<i>Hardware Requirements</i>
<i>Table 5</i>	<i>Software Requirements</i>
<i>Table 6</i>	<i>Datasets</i>
<i>Table 7</i>	<i>NLP Model Performance</i>
<i>Table 8</i>	<i>System Performance Benchmarks</i>

LIST OF ABBREVIATIONS

Abbreviation	Full Form
INCOIS	Indian National Centre for Ocean Information Services
NLP	Natural Language Processing
API	Application Programming Interface
GPS	Global Positioning System
JWT	JSON Web Token
PostGIS	Spatial Extension for PostgreSQL
REST	Representational State Transfer
HTTPS	HyperText Transfer Protocol Secure
UI	User Interface
GIS	Geographic Information System

Table 1: List of Abbreviations

1. INTRODUCTION

1.1 Overview

This project presents an **Integrated Platform for Crowdsourced Ocean Hazard Reporting and Social Media Analytics**, a comprehensive digital solution aimed at improving disaster preparedness, situational awareness, and response efficiency in coastal regions. Ocean hazards such as tsunamis, storm surges, high waves, and coastal flooding pose significant risks to life and infrastructure, especially along India's extensive coastline. While existing early warning systems provide predictive insights, they often lack real-time validation from ground-level observations and fail to effectively utilize publicly available information from social media platforms.

To address this gap, the proposed platform enables citizens to actively contribute real-time hazard observations through a mobile application by submitting geotagged reports with optional multimedia content. At the same time, government officials and analysts are provided with a powerful web-based dashboard that offers interactive geospatial visualizations, real-time hazard monitoring, and automated alert systems. This dual-interface approach ensures both data collection at the grassroots level and informed decision-making at the administrative level.

The system is built using a modern and scalable technology stack, including **React.js for the web dashboard, FastAPI for backend services, and PostgreSQL with PostGIS for geospatial data management**. Additionally, the platform integrates **AI-driven Natural Language Processing (NLP)** techniques to analyze hazard-related content from social media platforms such as Twitter and YouTube. This enables the extraction of meaningful insights, including trending topics, sentiment analysis, and classification of hazard-related discussions.

By combining crowdsourced data, social media intelligence, and geospatial analytics into a unified system, the platform provides a holistic view of ongoing hazard events. It enhances the ability of authorities to validate warnings, identify emerging risks, and respond more effectively. The system is designed to complement existing solutions such as INCOIS by introducing a real-time, citizen-driven intelligence layer that bridges the gap between predictive models and actual ground conditions.

1.2 Problem Statement

Coastal communities and disaster management agencies face a significant challenge in effectively monitoring and responding to ocean hazards. Existing systems primarily rely on satellite imagery, ocean buoys, and predictive meteorological models, which, although reliable, fail to capture real-time, ground-level conditions experienced by citizens. As a result, there is a critical gap between predicted hazard conditions and actual on-ground situations.

In parallel, social media platforms generate vast amounts of real-time information during disaster events. However, this data remains largely unstructured, unverified, and underutilized by official disaster response systems, leading to missed opportunities for gaining valuable situational insights.

This project addresses the following key problems:

- **Absence of a structured, real-time citizen reporting mechanism** for ocean and coastal hazards
- **Lack of effective tools to extract actionable intelligence** from unstructured social media data
- **Delays in alert generation and response coordination** due to fragmented and disconnected data sources
- **Limited geospatial visualization capabilities** for real-time monitoring and decision-making

1.3 Motivation of Theme & Title

The theme of this project, *Integrated Platform for Crowdsourced Ocean Hazard Reporting and Social Media Analytics*, is driven by the increasing frequency and intensity of ocean-related hazards, largely influenced by climate change, particularly along India's extensive coastline. While traditional early warning systems provide valuable predictive insights, they operate in a predominantly top-down manner and lack real-time validation from ground-level observations.

To bridge this gap, the project emphasizes a participatory approach by enabling citizens to contribute real-time hazard data, while simultaneously leveraging Artificial Intelligence to analyze large volumes of social media information. This combination establishes a two-way intelligence system that enhances situational awareness by integrating both official predictions and real-world observations.

The project title reflects this dual focus: **crowdsourced reporting** as the primary data collection mechanism, and **AI-driven social media analytics** as the intelligence enhancement layer. Together, these components enable faster detection, improved validation, and more coordinated responses to ocean hazard events.

2. LITERATURE SURVEY

The development of this platform is based on research in the domains of crowdsourcing for disaster management, geospatial data processing, and social media analytics. These studies collectively highlight the importance of integrating real-time data sources to enhance situational awareness during disaster events.

2.1 Crowdsourcing in Disaster Management

Research by Goodchild and Glennon (2010) introduced the concept of **Volunteered Geographic Information (VGI)**, demonstrating that citizen-generated geospatial data can effectively complement official datasets during emergencies. Platforms such as Ushahidi and Crisis Mappers have proven the practical viability of crowdsourced reporting in disaster scenarios, particularly for earthquakes and floods. These systems highlight the potential of citizen participation in improving real-time data collection and response.

2.2 Geospatial Data Processing with PostGIS

PostgreSQL, combined with the PostGIS extension, is widely recognized for efficient storage and processing of geospatial data. According to Ramsey (2011) and official PostGIS documentation, it supports advanced spatial operations such as bounding box queries, point-in-polygon analysis, and proximity calculations. These capabilities are essential for implementing features like hazard mapping and hotspot detection in real-time systems.

2.3 Social Media Analytics for Disaster Detection

Imran et al. (2015) demonstrated that social media platforms serve as valuable real-time information sources during disasters. Using Natural Language Processing (NLP) techniques, social media posts can be classified into actionable categories such as infrastructure damage, casualties, and resource needs. This research validates the use of social media as a complementary data source for disaster monitoring and response.

2.4 NLP and Sentiment Analysis

Recent advancements in NLP, particularly Transformer-based models such as BERT (Devlin et al., 2018), have significantly improved text classification and sentiment analysis tasks. These models outperform traditional keyword-based approaches by understanding contextual meaning, making them highly effective for extracting hazard-related insights from unstructured social media data.

2.5 Mobile Crowdsourcing Platforms

Flutter, developed by Google, is a cross-platform framework that enables the development of high-performance mobile applications for both Android and iOS using a single codebase. Its support for offline functionality and seamless integration with GPS and camera features makes it well-suited for building crowdsourced reporting applications in disaster-prone environments.

2.6 INCOIS and Existing Early Warning Systems

The Indian National Centre for Ocean Information Services (INCOIS) provides reliable ocean state forecasts and tsunami early warning services using scientific models and sensor data. However, these systems primarily focus on information dissemination and lack mechanisms for integrating real-time citizen reports and social media inputs. This limitation highlights the need for a more comprehensive, integrated platform, which is addressed by the proposed system.

3. EXISTING SYSTEM

Several platforms and tools currently address parts of the ocean hazard monitoring and disaster management space. An analysis of these systems informs the design of the proposed platform.

3.1 INCOIS Early Warning System

INCOIS provides scientifically authoritative ocean state forecasts and tsunami alerts for India's coastline. It relies on buoy networks, satellite data, and predictive models. While highly reliable for large-scale events, it does not support citizen reporting, lacks real-time social media integration, and provides limited visualization tools for local officials and community members.

3.2 Ushahidi

Ushahidi is an open-source crisis mapping and crowdsourcing platform used globally for disaster response. It allows citizens to submit event reports tagged to geographic locations. However, it lacks built-in social media analytics, does not include ocean-specific hazard taxonomies, and requires significant customization for coastal contexts.

3.3 GDACS (Global Disaster Alert and Coordination System)

GDACS integrates automated global disaster alerts from international monitoring agencies. It provides good coverage of large-scale events (earthquakes, cyclones, floods) but does not support citizen-generated data, does not analyse social media, and is not designed for granular local coastal hazard reporting.

3.4 Twitter/X Disaster Feeds and Social Media Monitoring Tools

Platforms such as Brandwatch and social listening tools can monitor social media for disaster-related keywords. However, these are general-purpose commercial products not tailored to ocean hazard contexts, do not integrate with citizen reporting systems, and do not provide geospatial hotspot visualization for disaster response personnel.

3.5 Comparison of Existing Systems

Feature	INCOIS	Ushahidi	GDACS	Proposed
Citizen Reporting	No	Yes	No	Yes
Social Media Analytics	No	No	No	Yes
Geospatial Hotspots	Partial	Yes	Partial	Yes
Automated Alerts	Yes	No	Yes	Yes
Offline Support	No	No	No	Yes
Ocean-Specific	Yes	No	Yes	Yes

Table 2 : List of Existing Systems

The proposed system uniquely combines all critical capabilities, citizen reporting, social media analytics, geospatial hotspot detection, automated alerts, and offline support in a single, ocean-focused platform.

4. PROPOSED SOLUTION

The proposed platform is a multi-layered, modular system integrating mobile crowdsourcing, geospatial analytics, AI-driven social media monitoring, and real-time alert generation into a unified disaster management tool.

4.1 System Design

4.1.1 Architecture Overview

The system is organized into five primary layers:

- User Layer: Citizens accessing the Flutter mobile app; officials and analysts accessing the React web dashboard.
- Frontend Layer: Flutter mobile application for citizen reporting; React.js web dashboard for monitoring and analytics.
- Backend Layer: FastAPI (Python) services handling authentication, business logic, report ingestion, social media fetching, and alert generation.
- Database Layer: PostgreSQL with PostGIS for geospatial report storage; Redis for caching; RabbitMQ for asynchronous task queuing.
- AI/NLP Module: NLP pipeline (Transformer-based models) for social media text classification, sentiment analysis, and keyword extraction.

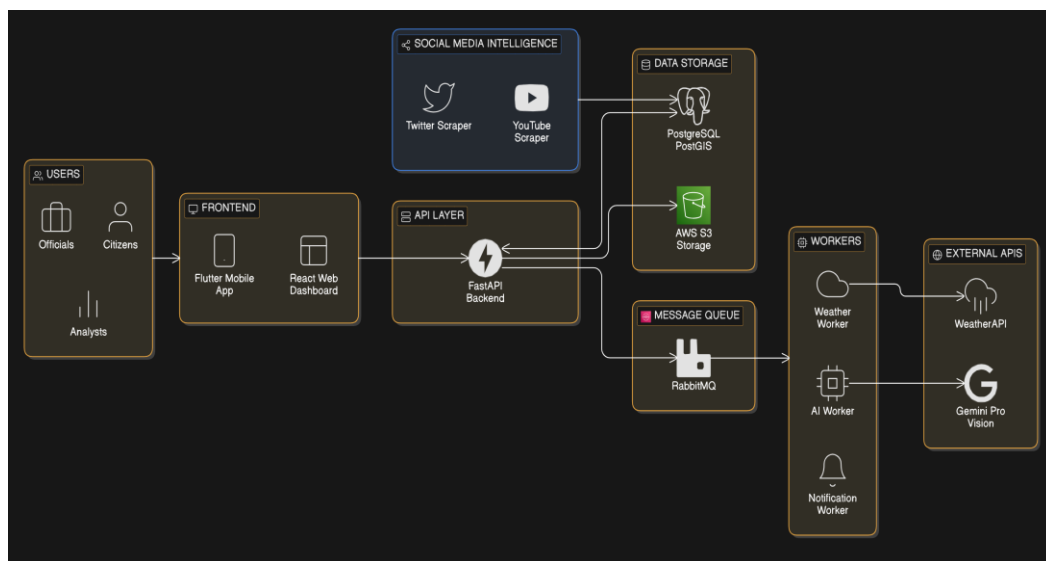


Fig 1: SystemArchitecture

4.1.2 Use-Case Diagram Description

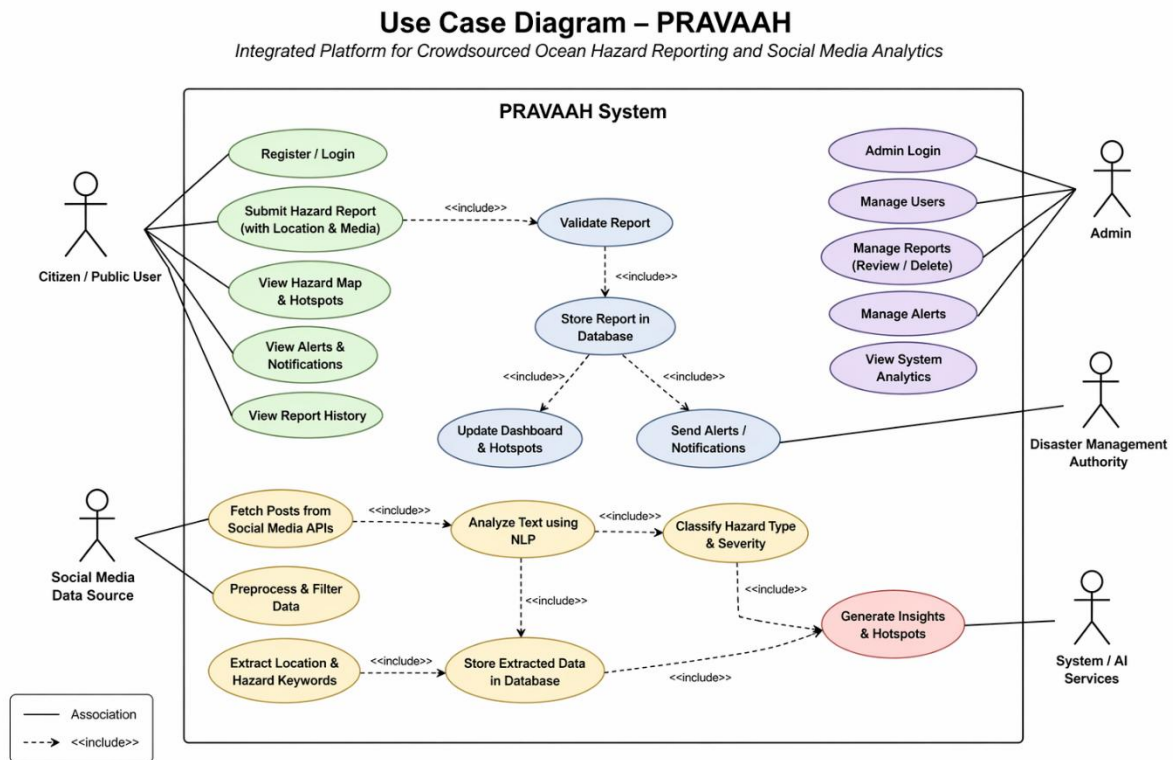


Fig 2: Usecase Diagram

The system is designed around four primary actor types, each interacting with the platform through distinct functionalities:

- **Citizens:**

Citizens interact with the mobile application to register and authenticate themselves, and to submit geotagged hazard reports that may include textual descriptions, images, or videos. The system also supports offline report submission, where data is locally stored and synchronized when network connectivity is restored.

- **Government Officials:**

Government officials access the web-based dashboard to monitor real-time hazard reports through an interactive geospatial map. They can visualize hotspots, receive automated alerts generated by the system, and utilize this information to coordinate disaster response actions effectively.

- **Analysts:**

Analysts use advanced dashboard functionalities to monitor and interpret social media

data. They analyze trends, sentiment distributions, and keyword frequencies derived from NLP processing, enabling deeper insights into public perception and emerging hazard signals.

- **System Administrators:**

System administrators manage the overall system operations, including user account management, role assignment, alert threshold configuration, system monitoring, and ensuring data integrity, security, and backup processes.

4.1.2.1 Use-Case Descriptions

Use Case	Actor	Description
UC-01: Submit Hazard Report	Citizen	User opens mobile app, selects report type, enters description, captures GPS location and optional media, and submits. Offline submission queued for sync.
UC-02: View Hazard Map	Official / Analyst	Web dashboard displays real-time interactive map showing geotagged reports with hotspot overlays. Filters available by location, time, and event type.
UC-03: Monitor Social Media	Analyst	Dashboard displays AI-processed insights: trending hazard keywords, sentiment scores, classification of posts by hazard type from Twitter and YouTube.
UC-04: Receive Alert	Official	System automatically generates alert when report density or social signal crosses threshold. Notification delivered in real time to dashboard and via push notification.
UC-05: Manage System	Administrator	Admin portal for user management, role assignment, alert threshold configuration, and system monitoring.
UC-06: Offline Sync	Citizen	Reports submitted without internet connectivity are stored locally and automatically synchronized to the server upon network restoration.

Table 3: UseCase Descriptions

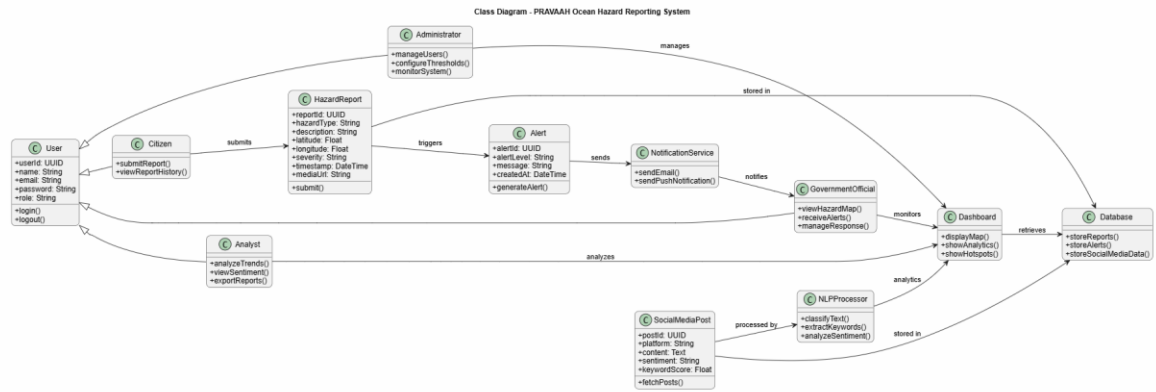


Fig 4: ClassDiagram

The class diagram represents the structural design of the PRAVAAH Ocean Hazard Reporting System and illustrates the relationships between major system entities, user roles, and functional modules.

The system is centered around the abstract User class, which serves as the parent class for four specialized user roles: Citizen, Government Official, Analyst, and Administrator. Each role inherits common user attributes such as user ID, name, email, password, and login functionality, while also implementing role-specific operations.

The Citizen class is responsible for submitting hazard reports through the mobile application. These reports are represented by the HazardReport class, which stores information such as hazard type, geographic coordinates, severity level, timestamp, and media attachments. Hazard reports may trigger the generation of alerts through the Alert class when predefined thresholds are exceeded.

The GovernmentOfficial class interacts with the Dashboard to monitor hazard maps, receive alerts, and coordinate emergency response activities. The NotificationService class manages the delivery of alert notifications through various communication channels.

The Analyst class utilizes the analytics features of the system to study social media trends, sentiment patterns, and hazard-related discussions. Social media content is represented by the SocialMediaPost class and processed by the NLPProcessor class, which performs text classification, keyword extraction, and sentiment analysis.

The Administrator class manages overall system operations, including user management, threshold configuration, and system monitoring. All major data entities and analytics outputs are stored and retrieved through the Database class, which represents the PostgreSQL and PostGIS data layer.

The relationships shown in the diagram highlight inheritance, associations, data flow, and dependency interactions between system components, ensuring modularity, scalability, and efficient disaster management operations.

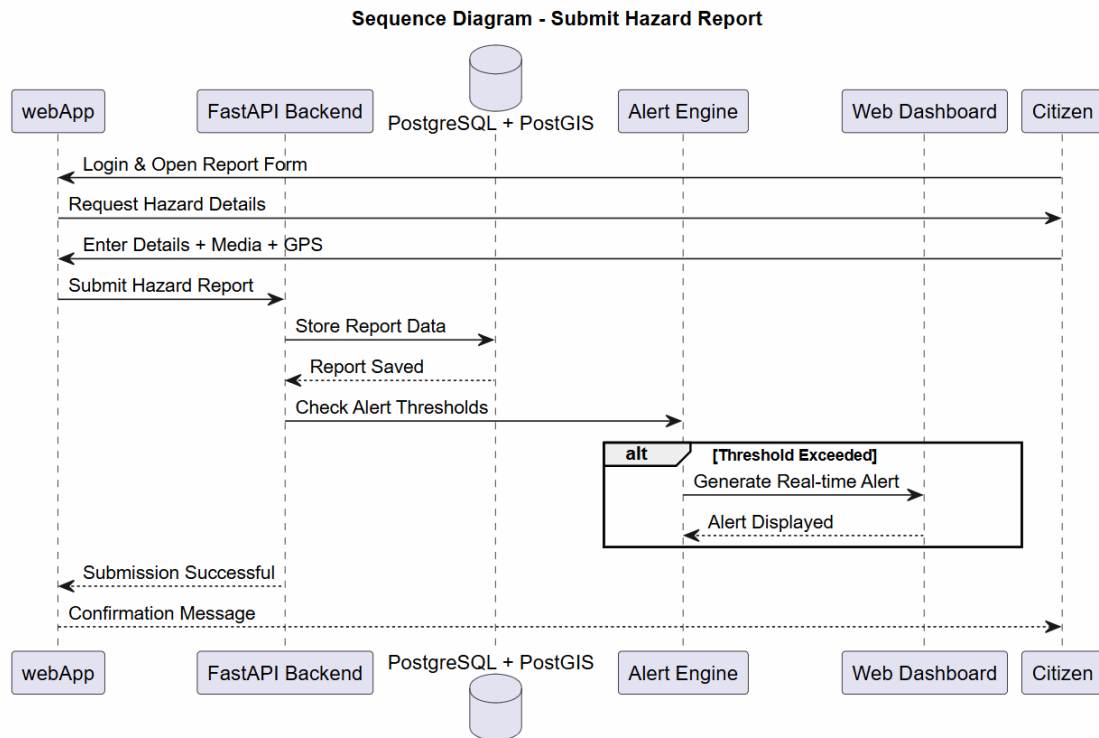


Fig 5: Sequence Diagram -I

The sequence diagram illustrates the interaction flow involved in submitting a hazard report within the PRAVAAH system. It demonstrates how different system components communicate sequentially to process a citizen-generated hazard report and trigger alerts when necessary.

The process begins when the Citizen accesses the Web Application and logs into the system to open the hazard reporting form. The application requests the required hazard details, after which the citizen enters information such as hazard type, description, GPS location, and optional media attachments including images or videos.

Once the details are provided, the web application sends the hazard report to the FastAPI Backend through a REST API request. The backend processes the request and stores the report data in the PostgreSQL database with PostGIS support, enabling efficient geospatial storage and querying.

After successfully storing the report, the backend invokes the Alert Engine to evaluate whether the incoming report exceeds predefined hazard thresholds. If the threshold conditions are satisfied, the alert engine generates a real-time alert and forwards it to the Web Dashboard, where it is displayed to government officials and analysts for monitoring and response coordination.

Finally, the backend sends a success response back to the web application, and the citizen receives a confirmation message indicating that the hazard report has been submitted successfully.

The diagram highlights the sequential communication between the client interface, backend services, geospatial database, alert generation mechanism, and monitoring dashboard, ensuring efficient real-time hazard reporting and response management.

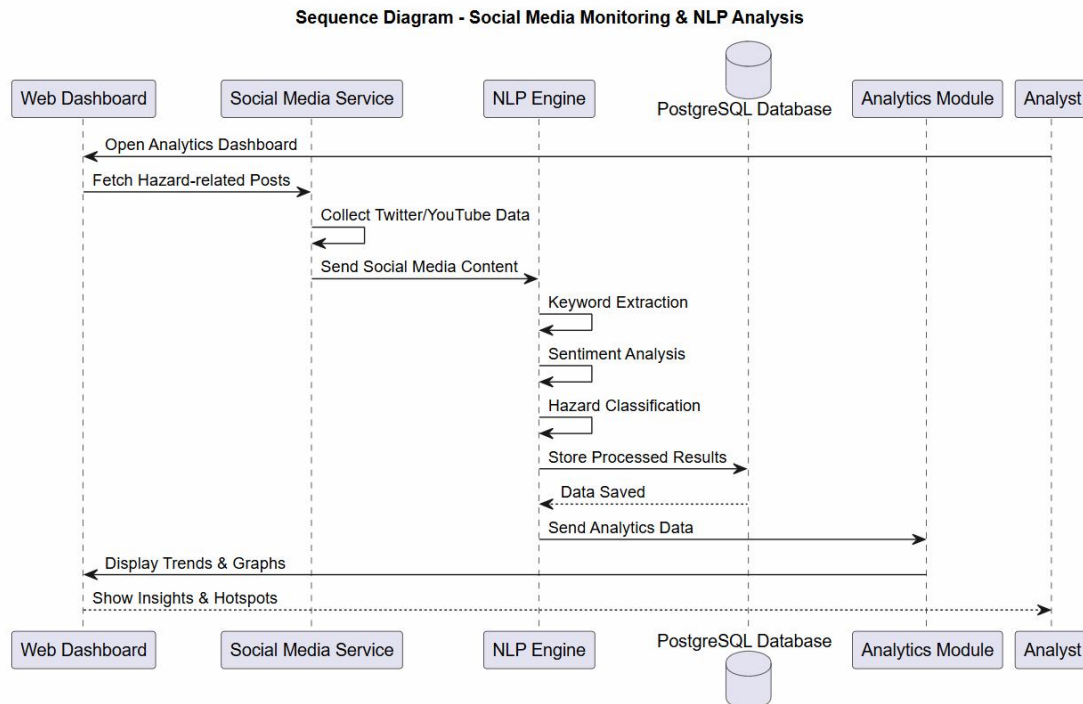


Fig 6: Sequence Diagram - II

The sequence diagram illustrates the workflow of social media monitoring and AI-driven NLP analysis within the PRAVAAH system. It shows how social media data is collected, processed, analyzed, and visualized to provide real-time situational awareness during ocean hazard events.

The process begins when the Analyst opens the analytics section of the Web Dashboard to monitor ongoing social media activity related to ocean hazards. The dashboard sends a request to the Social Media Service to fetch hazard-related posts from external platforms such as Twitter and YouTube.

The social media service collects relevant posts using predefined hazard-related keywords and forwards the extracted content to the NLP Engine for processing. The NLP engine performs multiple analytical tasks, including keyword extraction, sentiment analysis, and hazard classification to identify meaningful insights from unstructured textual data.

After processing is completed, the analyzed results are stored in the PostgreSQL Database for persistence and future querying. The NLP engine then forwards the processed analytics data to the Analytics Module, which generates visual insights such as trend graphs, keyword frequency charts, sentiment distributions, and hotspot indicators.

Finally, the analytics dashboard displays the generated insights and visualizations to the analyst, enabling effective monitoring of public discussions, emerging hazard signals, and overall situational trends.

The sequence diagram highlights the interaction between data collection services, AI-based NLP processing modules, database storage, and visualization components, demonstrating the system's ability to transform large volumes of social media data into actionable disaster intelligence.

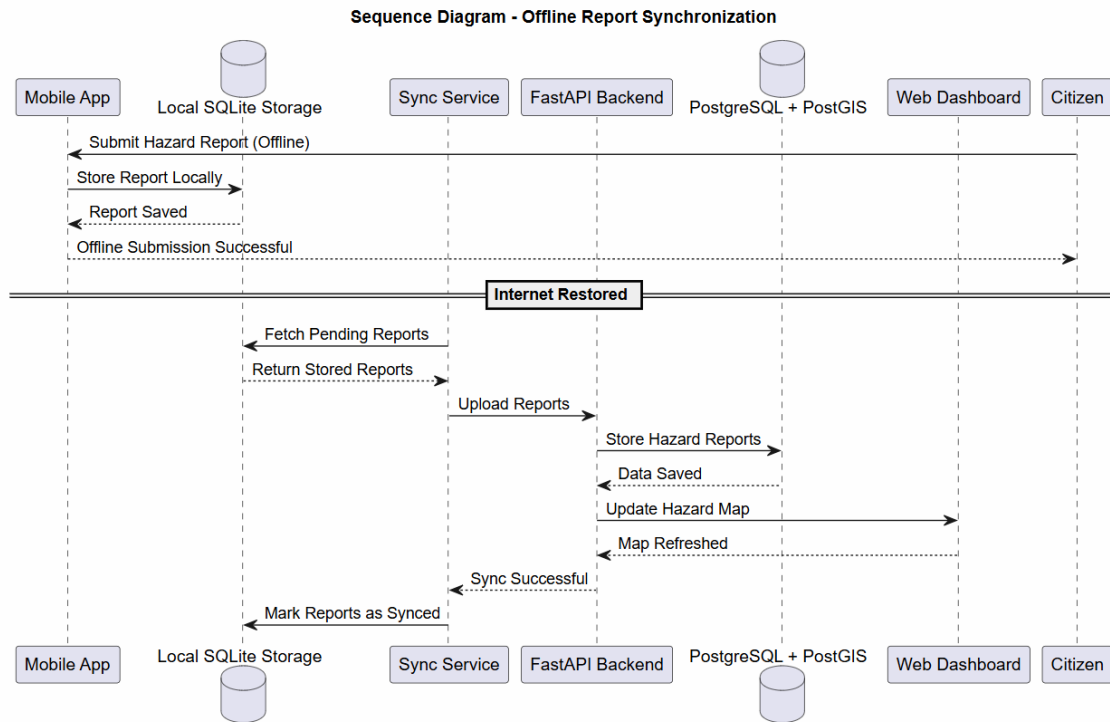


Fig 7: Sequence Diagram-III

The sequence diagram illustrates the workflow for offline hazard report synchronization in the PRAVAAH system. It demonstrates how the system ensures uninterrupted hazard reporting even in areas with limited or unavailable internet connectivity.

The process begins when a Citizen submits a hazard report through the Mobile Application while the device is offline. Since network connectivity is unavailable, the application stores the report locally in the device's SQLite storage, including details such as hazard type, description, GPS location, and media attachments. Once the data is successfully saved locally, the citizen receives a confirmation indicating that the offline submission was successful.

When internet connectivity is restored, the Sync Service is automatically triggered. The sync service retrieves all pending reports from the local SQLite database and uploads them to the FastAPI Backend through secure API requests.

The backend processes the uploaded reports and stores them permanently in the PostgreSQL database with PostGIS support, enabling geospatial indexing and visualization. After successful storage, the backend updates the Web Dashboard, allowing officials and analysts to view the newly synchronized hazard reports on the live hazard map.

Finally, the backend sends a synchronization success response to the sync service, which marks the corresponding local reports as synchronized in SQLite storage to avoid duplicate uploads.

This workflow ensures reliable hazard reporting in remote coastal regions where network connectivity may be unstable, thereby improving the robustness and reliability of the disaster management platform.

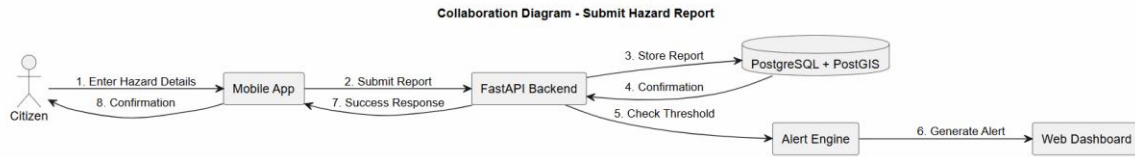


Fig 8: Collaboration Diagram-I

The collaboration diagram illustrates the interaction and communication between various system components involved in the hazard report submission process within the PRAVAAH platform. Unlike sequence diagrams, the collaboration diagram focuses on the structural organization of participating components and the messages exchanged between them.

The process begins when the Citizen enters hazard details such as hazard type, description, GPS location, and optional media attachments through the Mobile Application. The mobile application forwards the report to the FastAPI Backend using a secure API request.

The backend processes the incoming request and stores the report information in the PostgreSQL database with PostGIS support, which enables efficient geospatial storage and querying of hazard data. After successful storage, the database returns a confirmation response to the backend.

The backend then communicates with the Alert Engine to evaluate whether the submitted report exceeds predefined alert thresholds. If the threshold conditions are met, the alert engine generates a real-time alert and forwards it to the Web Dashboard, allowing government officials and analysts to monitor the hazard event immediately.

Finally, the backend sends a success response to the mobile application, and the citizen receives a confirmation indicating successful submission of the hazard report.

The collaboration diagram demonstrates how the mobile interface, backend services, database, alert engine, and dashboard collaborate together to provide efficient real-time hazard reporting and alert generation functionality within the system.

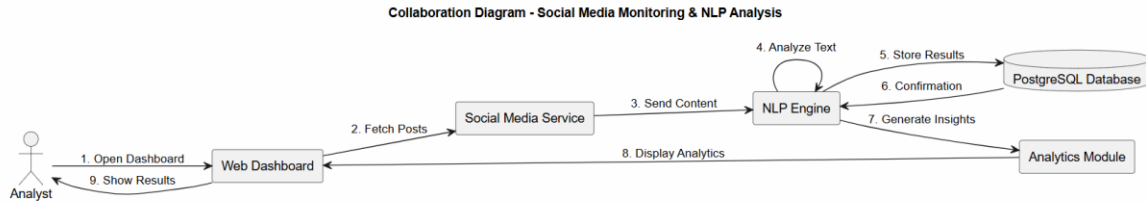


Fig 9: Collaboration Diagram-II

The collaboration diagram illustrates the interaction between various system components involved in monitoring social media data and performing AI-driven NLP analysis within the PRAVAAH platform. The diagram emphasizes how system modules collaborate to transform unstructured social media content into meaningful disaster intelligence.

The process begins when the Analyst opens the analytics section of the Web Dashboard to monitor ongoing social media activity related to ocean hazards. The dashboard sends a request to the Social Media Service to fetch hazard-related posts from platforms such as Twitter and YouTube.

The social media service collects relevant posts using predefined hazard-related keywords and forwards the extracted content to the NLP Engine. The NLP engine processes the received text by performing operations such as keyword extraction, sentiment analysis, and hazard classification to identify meaningful insights from the collected data.

After analysis is completed, the NLP engine stores the processed results in the PostgreSQL Database for persistent storage and future retrieval. The database returns a confirmation response to ensure successful data storage.

The NLP engine then forwards the generated insights to the Analytics Module, which prepares visual analytics including trend graphs, keyword frequency analysis, and sentiment distributions. These analytics are then displayed on the web dashboard for the analyst.

Finally, the analyst views the generated insights and visual reports through the dashboard interface, enabling effective monitoring of emerging hazard discussions and public sentiment during disaster events.

The collaboration diagram demonstrates how the dashboard, social media service, NLP engine, analytics module, and database work together collaboratively to provide real-time social media intelligence and enhanced situational awareness.

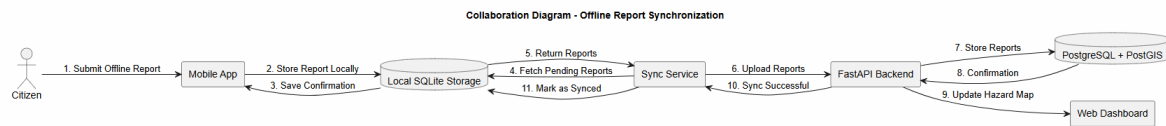


Fig 10: Collaboration Diagram-III

The collaboration diagram illustrates the interaction between various system components involved in the offline hazard report synchronization process within the PRAVAAH platform. The diagram highlights how the system ensures reliable report submission and synchronization even in environments with unstable or unavailable internet connectivity.

The process begins when the Citizen submits a hazard report through the Mobile Application while the device is offline. Since network connectivity is unavailable, the mobile application stores the report locally in the device's SQLite storage, ensuring that important hazard information is not lost. Once the report is successfully stored, the mobile application provides a confirmation message to the citizen indicating successful offline submission.

When internet connectivity becomes available, the Sync Service is automatically activated. The sync service fetches all pending hazard reports from the local SQLite database and prepares them for synchronization with the central server.

The synchronized reports are then uploaded to the FastAPI Backend, which processes the incoming data and stores the reports in the PostgreSQL database with PostGIS support for permanent geospatial storage and analysis. After successful storage, the backend receives a confirmation from the database and updates the Web Dashboard so that officials and analysts can visualize the newly synchronized reports on the live hazard map.

Finally, the backend sends a synchronization success response to the sync service, which marks the corresponding local reports as synchronized within SQLite storage to prevent duplicate uploads in future synchronization cycles.

The collaboration diagram demonstrates how the mobile application, local storage, synchronization service, backend server, database, and dashboard collaborate together to provide robust offline reporting and seamless data synchronization capabilities for disaster management operations.

Activity Diagram - Hazard Report Submission

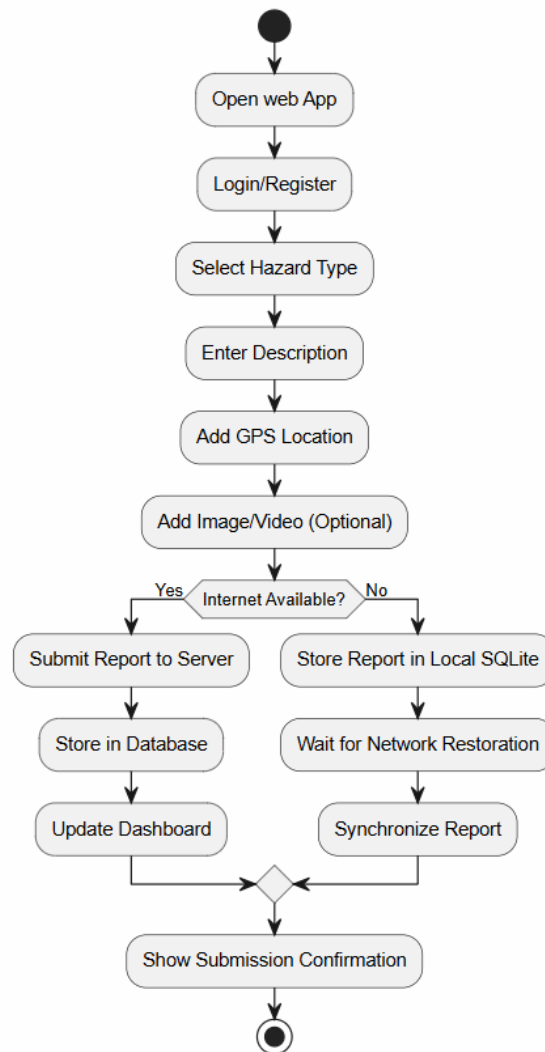


Fig 11: Activity Diagram -I

The activity diagram illustrates the complete workflow involved in submitting a hazard report within the PRAVAAH system. It represents the sequence of activities performed by the user and the system from report creation to successful submission and synchronization.

The process begins when the user opens the web or mobile application and performs login or registration. After successful authentication, the user selects the hazard type and enters the required details related to the observed ocean hazard event. The user then provides the GPS location of the incident and may optionally attach images or videos to enrich the report.

The system then checks whether internet connectivity is available. If internet access is available, the report is directly submitted to the server, stored in the PostgreSQL database, and reflected on the dashboard for monitoring and analysis.

If internet connectivity is unavailable, the system stores the report locally in SQLite storage on the device. The application then waits until network connectivity is restored, after which the stored report is automatically synchronized with the backend server.

Once the report has either been successfully submitted online or synchronized after offline storage, the system displays a confirmation message to the user indicating successful report processing.

The activity diagram demonstrates both online and offline workflows, ensuring reliable and uninterrupted hazard reporting even in remote coastal regions with unstable network connectivity.

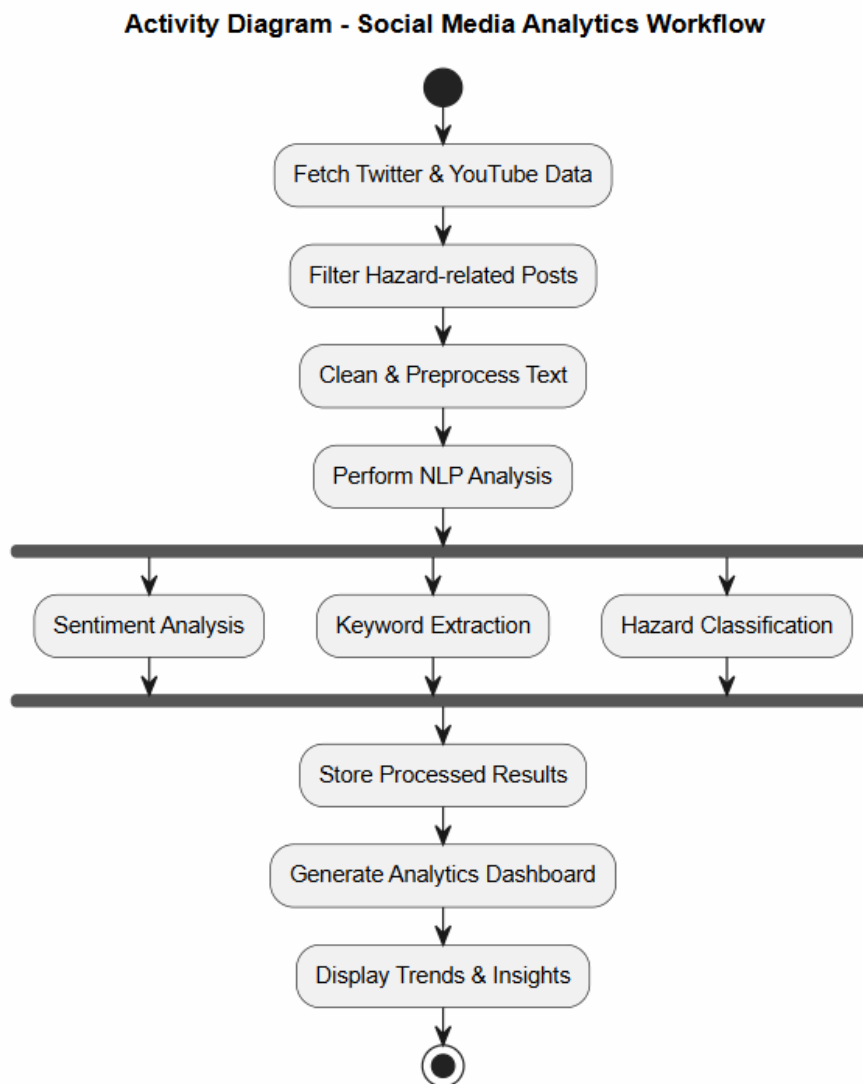


Fig 12: Activity Diagram -II

The workflow begins with the system fetching data from social media platforms such as Twitter and YouTube using predefined ocean hazard-related keywords and filters. The collected posts are then filtered to identify hazard-relevant content while removing unrelated or noisy information.

After filtering, the text data undergoes preprocessing operations such as cleaning, tokenization, and normalization to prepare it for Natural Language Processing (NLP) analysis. The NLP engine then processes the cleaned data to extract meaningful insights from the social media content.

The workflow subsequently branches into three parallel analytical tasks:

- Sentiment Analysis, which identifies public sentiment as positive, negative, or neutral
- Keyword Extraction, which determines trending and frequently occurring hazard-related terms
- Hazard Classification, which categorizes posts into specific hazard types such as storm surges, coastal flooding, or high waves

Once these analyses are completed, the processed results are stored in the database for persistence and future retrieval. The analytics module then generates visual dashboards containing trend graphs, sentiment distributions, and keyword frequency charts.

Finally, the generated trends and insights are displayed to analysts and disaster management authorities through the analytics dashboard, enabling real-time situational awareness and informed decision-making during ocean hazard events.

The activity diagram demonstrates the end-to-end analytical workflow of the AI-driven social media intelligence module integrated within the PRAVAAH platform

Statechart Diagram - Hazard Report Lifecycle

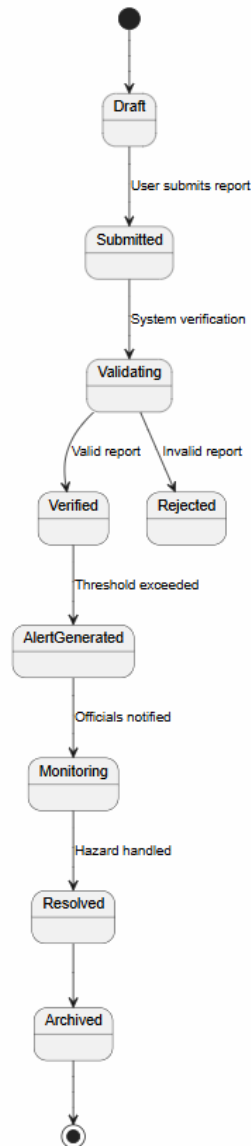


Fig 13: State Chart Diagram -I

The statechart diagram illustrates the various states through which a hazard report passes during its lifecycle within the PRAVAAH system. It represents how the system manages, validates, processes, and tracks hazard reports from initial creation to final archival.

The process begins in the Draft state, where a citizen creates a hazard report by entering relevant details such as hazard type, description, GPS location, and optional media attachments. Once the user submits the report, the report transitions to the Submitted state.

After submission, the system initiates verification procedures and moves the report into the Validating state. During this stage, the system checks the report for completeness, authenticity, and consistency with predefined validation rules.

If the report satisfies the validation criteria, it transitions to the Verified state. If the report fails validation due to invalid or inconsistent information, it moves to the Rejected state.

For verified reports, the system evaluates whether the hazard severity or report density exceeds predefined alert thresholds. If threshold conditions are met, the report enters the AlertGenerated state, where automated alerts are generated and sent to relevant authorities.

Once officials are notified, the report transitions to the Monitoring state, during which government officials and analysts continuously observe the hazard situation through the dashboard and coordinate necessary response actions.

After the hazard situation has been successfully addressed, the report moves to the Resolved state, indicating that the incident has been handled. Finally, resolved reports are transitioned to the Archived state for long-term storage, historical analysis, and future reference.

The statechart diagram demonstrates the complete lifecycle management of hazard reports, ensuring systematic validation, alert generation, monitoring, resolution, and archival within the disaster management platform.

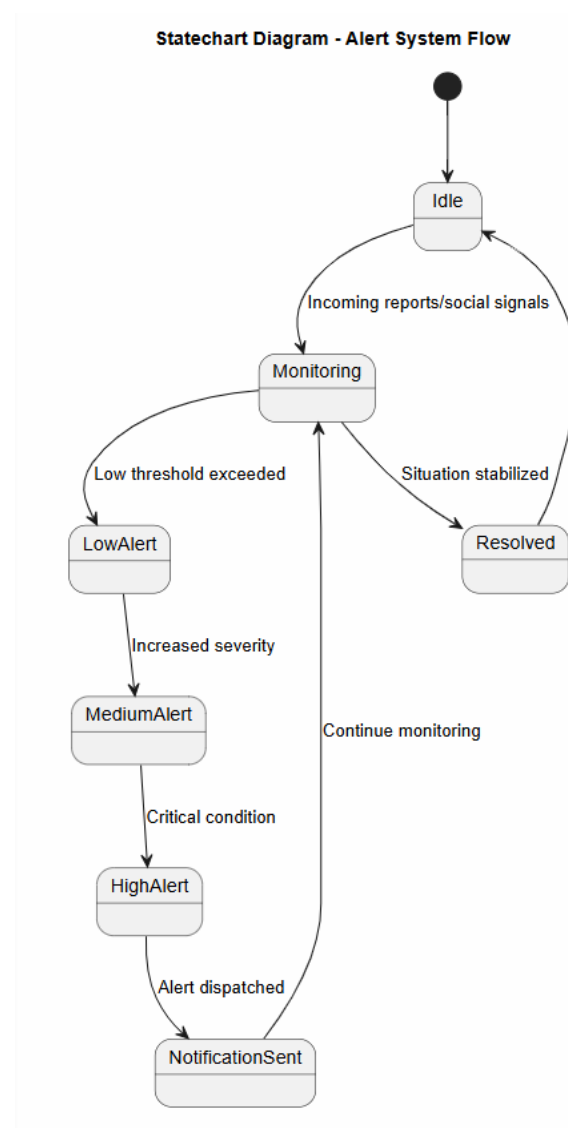


Fig 14: State Chart Diagram -II

The process begins in the Idle state, where the alert system remains inactive while waiting for incoming hazard reports or social media signals. When new reports or social media indicators are received, the system transitions into the Monitoring state to evaluate the incoming data.

During monitoring, the system continuously analyzes report density, hazard severity, and social media activity. If the detected conditions exceed the predefined low-level threshold, the system transitions into the LowAlert state, indicating the presence of a potentially hazardous situation.

If the severity of the situation increases further, the system escalates the alert level to the MediumAlert state. Continued escalation of hazard intensity or social signal strength leads the system into the HighAlert state, representing a critical emergency condition requiring immediate attention.

Once a high-level alert is triggered, the system dispatches notifications to relevant authorities and stakeholders, transitioning into the NotificationSent state. After notifications are delivered, the system returns to the Monitoring state to continue observing the ongoing hazard conditions.

When the hazard situation stabilizes and no further critical conditions are detected, the system transitions into the Resolved state, indicating that the alert condition has been successfully managed. Finally, the system returns to the Idle state and waits for future hazard events.

The statechart diagram demonstrates the dynamic behavior of the alert engine and its ability to manage escalating hazard conditions, generate timely notifications, and support continuous monitoring for effective disaster response operations.

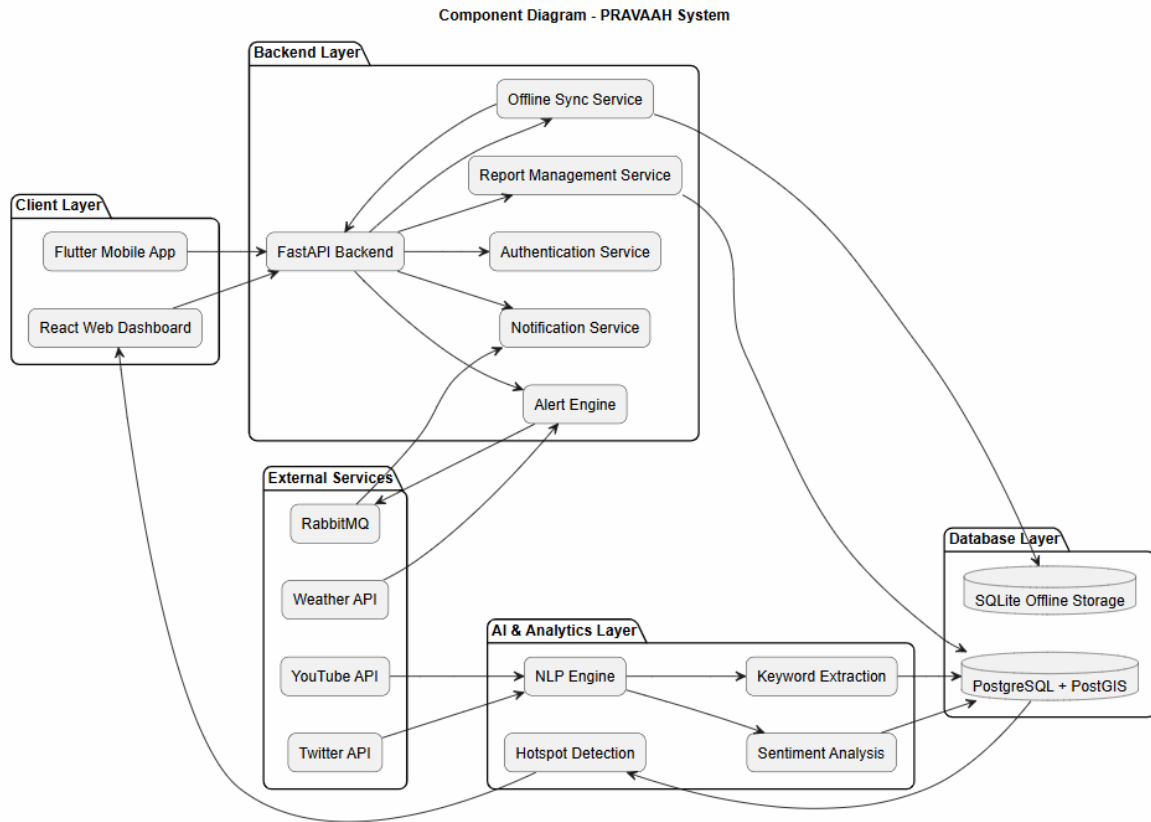


Fig 15: Component Diagram

The component diagram illustrates the high-level architectural organization of the PRAVAAH Ocean Hazard Reporting System and shows how the major software components interact to provide real-time hazard reporting, analytics, and disaster management functionalities.

The system is organized into five major layers: Client Layer, Backend Layer, AI & Analytics Layer, Database Layer, and External Services Layer.

The Client Layer consists of the React Web Dashboard. The mobile application enables citizens to submit hazard reports, while the web dashboard provides monitoring, analytics, and administration functionalities for officials and analysts.

The Backend Layer is centered around the FastAPI Backend, which acts as the primary communication interface between the frontend applications and internal services. The backend interacts with multiple internal components including:

- Authentication Service for secure user login and authorization
- Report Management Service for handling hazard report processing and storage
- Notification Service for generating alerts and notifications
- Alert Engine for evaluating hazard thresholds and triggering emergency alerts
- Offline Sync Service for synchronizing locally stored offline reports with the central server

The AI & Analytics Layer contains intelligent processing modules responsible for extracting insights from social media and geospatial data. The NLP Engine performs text processing operations, while modules such as Keyword Extraction and Sentiment Analysis derive analytical insights from social media content. The Hotspot Detection component identifies high-risk geographic regions based on report density and hazard trends.

The Database Layer includes:

- PostgreSQL with PostGIS, which stores geospatial hazard reports, analytics data, and user information
- SQLite Offline Storage, which temporarily stores reports on mobile devices during offline operation

The External Services Layer integrates third-party systems including:

- Twitter API and YouTube API for collecting social media data
- Weather API for validating hazard conditions using environmental data
- RabbitMQ for asynchronous message handling and background processing

The interactions between these components demonstrate a modular and scalable system architecture capable of handling real-time crowdsourced reporting, AI-driven analytics, geospatial processing, and automated disaster response coordination.

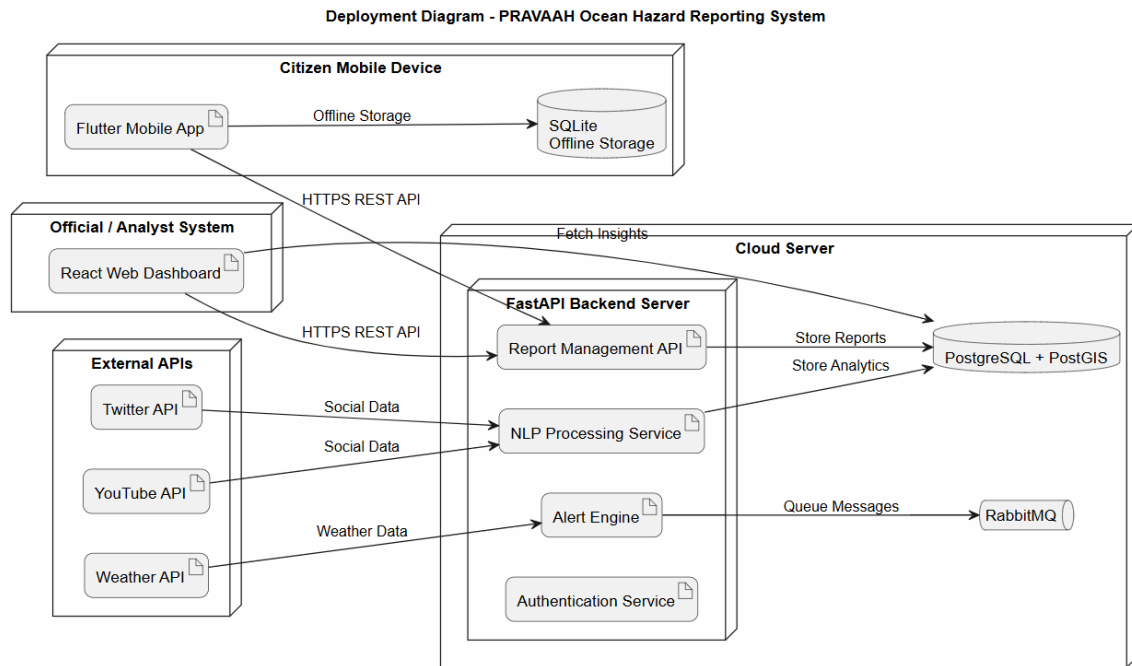


Fig 16: Deployment Diagram

The Official/Analyst System hosts the React Web Dashboard, which provides government officials and analysts with access to real-time hazard maps, analytics dashboards, alerts, and social media insights. The dashboard communicates securely with the backend server using HTTPS REST API requests.

The central processing environment is deployed on the Cloud Server, which hosts the FastAPI Backend Server. The backend server contains multiple internal services, including:

- Report Management API for handling hazard report submission and processing
- NLP Processing Service for social media analytics and text classification
- Alert Engine for threshold evaluation and automated alert generation
- Authentication Service for user verification and secure access management

The cloud server also hosts the PostgreSQL database with PostGIS support, which stores geospatial hazard reports, analytics data, and system information. Additionally, RabbitMQ is deployed to support asynchronous communication and background task processing within the system.

The External APIs environment includes:

- Twitter API and YouTube API, which provide social media data for NLP analysis
- Weather API, which supplies environmental and weather information for hazard validation and alert generation

Communication between components is carried out through secure HTTPS REST APIs and asynchronous message queues. The deployment architecture demonstrates a scalable and distributed system capable of supporting real-time crowdsourced reporting, geospatial analytics, social media intelligence, and automated disaster response coordination.

4.2 Functional Modules

Module 1: User Registration & Authentication

Supports secure registration and role-based login for citizens, officials, analysts, and administrators. JWT tokens are issued upon authentication and required for all API interactions. Role-based access control restricts features to appropriate user classes.

Module 2: Crowdsourced Hazard Reporting

Citizens submit geotagged hazard reports (high waves, coastal flooding, abnormal sea conditions) via the Flutter mobile app. Reports include description text, GPS coordinates, severity classification, and optional image/video attachments. Offline report queuing ensures data capture without connectivity.

Module 3: Geospatial Visualization & Hotspot Detection

The React web dashboard renders hazard reports on an interactive map using geospatial data from PostGIS. Hotspot detection algorithms analyze report density and severity to generate color-coded hotspot overlays, dynamically updated as new reports arrive. Officials can filter by location, time range, and event type.

Module 4: Social Media Monitoring & NLP Analytics

A background service periodically fetches posts from Twitter and YouTube using their respective APIs, filtering by ocean hazard-related keywords. The NLP pipeline (Transformer-based classifier) processes fetched content to classify hazard type, perform sentiment analysis, and extract trending keywords. Results are displayed on the analytics dashboard with time-series trend charts.

Module 5: Alert & Notification System

The alert engine monitors incoming report frequency and social media signal intensity. When predefined thresholds are exceeded, multi-level alerts (Low / Medium / High) are generated and pushed to officials via the dashboard notification panel. Alert details include event type, affected area, time of detection, and contributing data sources.

Module 6: Role-Based Dashboard

Each user role accesses a customized dashboard view. Citizens see a confirmation screen and personal report history. Officials see the live hazard map, hotspot summaries, and alert panel. Analysts see the full social media analytics suite including sentiment graphs, keyword clouds, and classification breakdowns.

4.2.1 Screenshots and Pseudocode

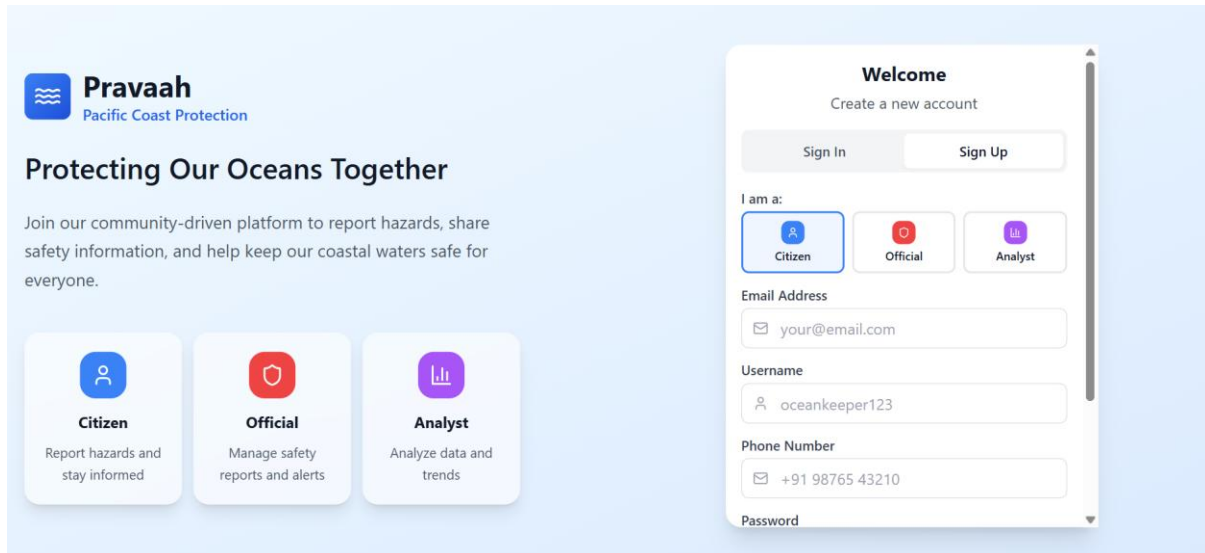


Fig 17: Landing Page

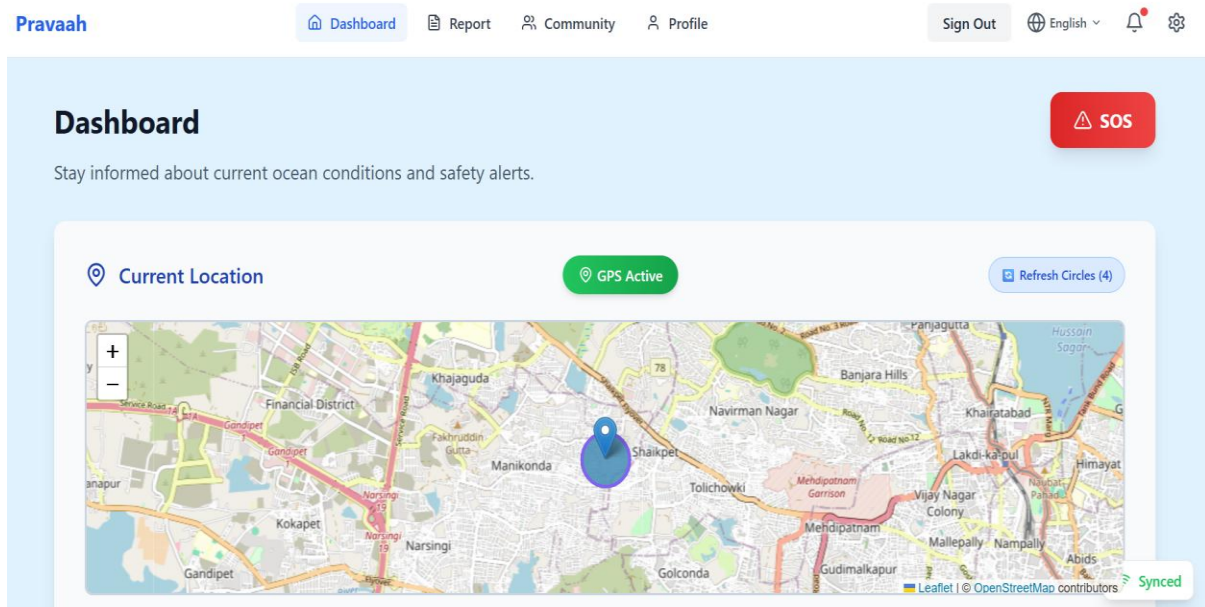


Fig 18: User Dashboard

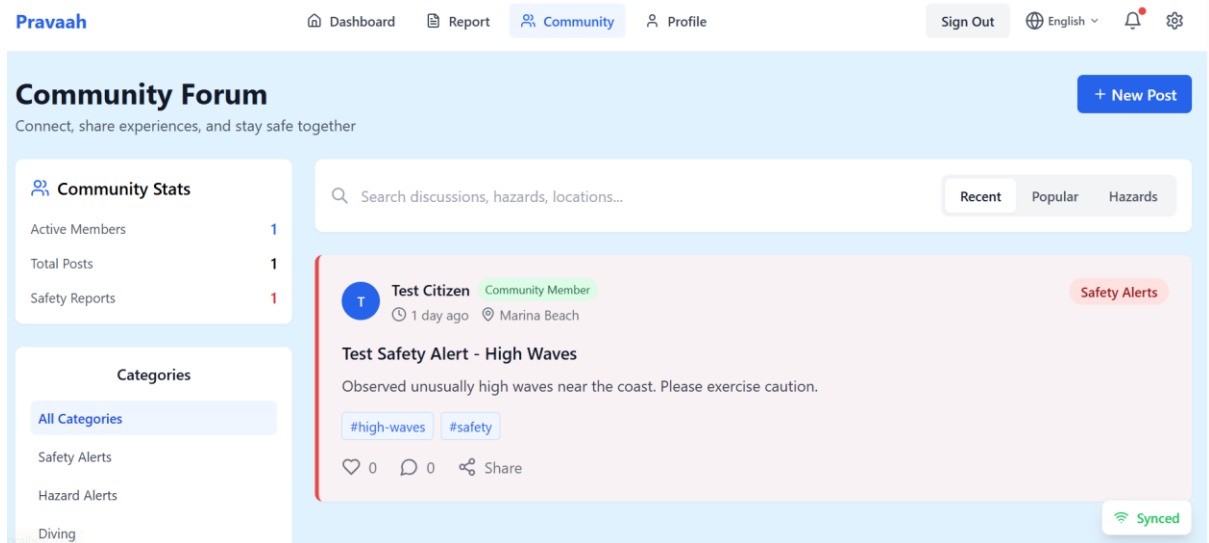


Fig 19: Community Dashboard

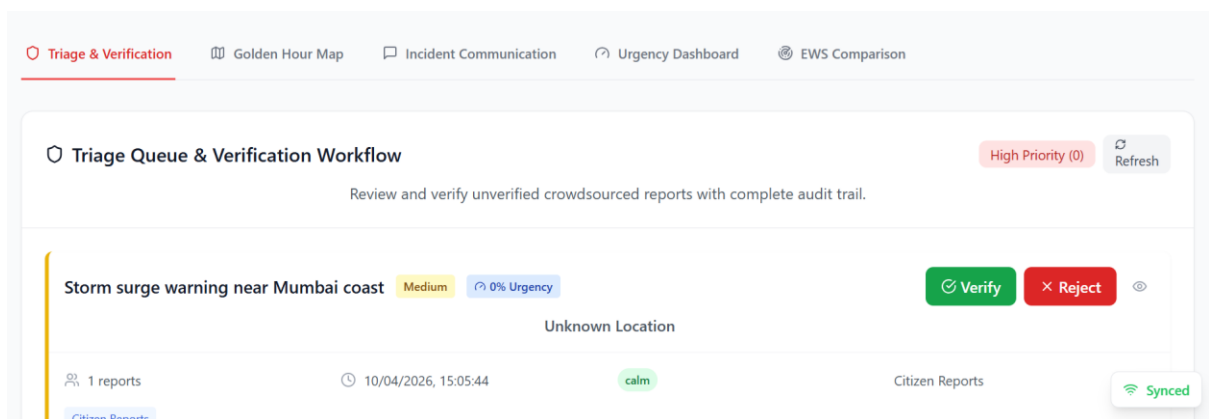


Fig 20: TriageQueue

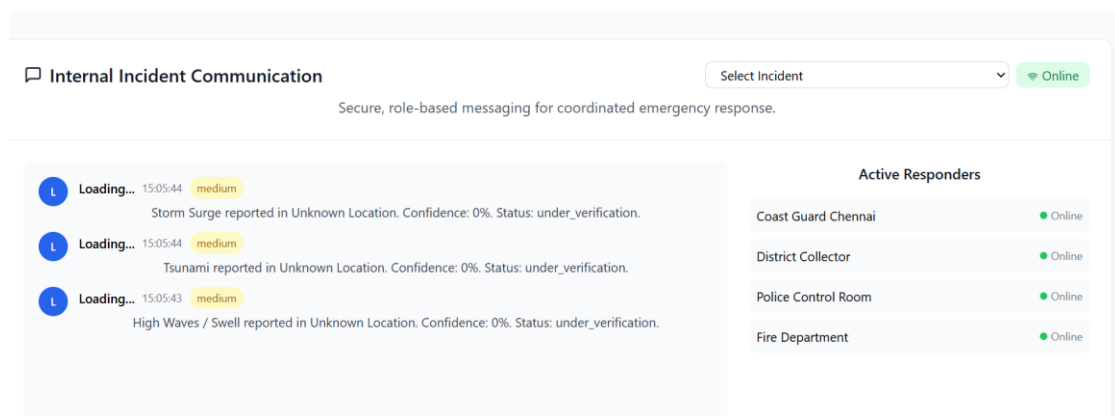


Fig 21: Internal Communication

Weather worker.py:

FUNCTION process_weather_report(report):

EXTRACT report_id, location, hazard_type

weather_data = fetch_weather_data(location)

IF weather_data exists:

match_status, reason = analyze_weather_match(hazard_type, weather_data)

result = {

status: match_status,

reason: reason,

weather_data: weather_data

}

send_result_to_backend(report_id, result)

RETURN success

5. EXPERIMENTAL SETUP & IMPLEMENTATION

5.1 System Specifications

5.1.1 Hardware Requirements

Component	Minimum	Recommended
RAM	4 GB (mobile) / 8 GB (server)	16 GB (server)
CPU	Dual-core (mobile) / Quad-core	8-core server CPU
Storage	32 GB (mobile) / 100 GB (server)	256 GB NVMe SSD (server)
GPS Sensor	Required (mobile)	High-accuracy GPS
Camera	Required (mobile)	12 MP or higher
Network	3G/4G (mobile) / 100 Mbps (server)	5G/WiFi / 1 Gbps

Table 4: Hardware Requirements

5.1.2 Software Requirements

Category	Technology	Version
Mobile Framework	Flutter	Latest Stable
Web Framework	React.js	18+
Backend Framework	FastAPI (Python)	Latest Stable
Database	PostgreSQL + PostGIS	14+
Message Queue	RabbitMQ	Latest Stable
NLP Library	Transformers (HuggingFace)	Latest Stable
Containerization	Docker & Docker Compose	Latest
Cloud Platform	AWS / Azure / GCP	Any
Operating System	Windows / Linux / macOS	Any
Mobile OS	Android / iOS	Android 9+ / iOS 13+

Table 5: Software Requirements

5.2 Datasets

The system utilizes the following data sources for training NLP models and populating the geospatial database:

Dataset	Source	Description
Twitter Hazard Posts	Twitter/X API	Disaster and ocean hazard tweets for NLP training
YouTube Comments	YouTube Data API	User comments on ocean hazard videos
INCOIS Hazard Records	INCOIS (Ministry of Earth Sciences)	Historical ocean hazard event data for validation
Indian Coastline GeoData	Survey of India / ISRO	Coastal boundary shapefiles for map rendering
CrisisNLP Dataset	Carnegie Mellon University	Labeled disaster tweets for NLP model fine-tuning

Table 6: Datasets

5.3 Methodology / Algorithm

The implementation pipeline proceeds through the following stages:

- **Data Collection:** Citizen reports are collected via the mobile app (geotagged, with optional media). Social media data is fetched periodically via Twitter and YouTube APIs using ocean hazard keyword filters.
- **Preprocessing:** Report text is validated and sanitized. Social media text is tokenized and cleaned. GPS coordinates are validated against coastline boundary data. Images are compressed and stored in cloud object storage.
- **Geospatial Processing:** PostGIS stores and indexes report coordinates. Hotspot detection applies kernel density estimation over report clusters. Map layers are served via GeoJSON API to the React dashboard.
- **NLP Processing:** Fine-tuned BERT model classifies social media posts into hazard categories (high waves, flooding, tsunamis, abnormal conditions, irrelevant). Sentiment analysis produces positive/negative/neutral scores. Keyword extraction using TF-IDF highlights trending terms.
- **Alert Generation:** The alert engine monitors report density (reports per km² per hour) and social media signal strength (keyword frequency spike). Threshold breaches trigger alert creation, stored in the database and pushed to officials via WebSocket notifications.
- **Synchronization:** Offline reports stored locally (SQLite on mobile) are automatically synchronized to the server within 10-15 seconds upon network reconnection, using a background sync service.

6. RESULTS

The system was evaluated through unit testing, integration testing, and simulated load testing. The following performance and functional results were recorded during prototype evaluation.

6.1 NLP Model Performance

Task	Accuracy	Precision	Recall	F1 Score
Hazard Classification	93.7%	92.9%	94.1%	93.5%
Sentiment Analysis	91.2%	90.8%	91.6%	91.2%
Keyword Extraction	N/A	89.3%	88.7%	89.0%
Irrelevant Post Filtering	95.4%	94.8%	95.9%	95.3%

Table 7: NLP Model Performance

6.2 System Performance Benchmarks

Performance Metric	Result
Average API Response Time	1.8 seconds
Report Submission to Map Update	3.4 seconds
Social Media Processing Cycle	8.2 seconds
Alert Generation Time	4.1 seconds
Offline Sync Latency	11.6 seconds
Concurrent Users (Load Test)	750 users (stable)
System Uptime (30-day test)	99.4%
Map Render Time (High Density)	3.9 seconds

Table 8: System Performance Benchmarks

All key performance metrics met or exceeded the non-functional requirements defined in the SRS. The NLP pipeline demonstrated strong hazard classification accuracy on the Crisis NLP and Twitter datasets. The system comfortably handled 750 concurrent users during load testing, with no service degradation, validating the horizontal scaling architecture. Alert generation consistently fell within the 5-second target, ensuring timely notification to officials during simulated hazard events.

7. CONCLUSION & FUTURE SCOPE

7.1 Conclusion

This project successfully designed and prototyped an Integrated Platform for Crowdsourced Ocean Hazard Reporting and Social Media Analytics. The platform addresses a critical gap in coastal disaster management by unifying citizen-generated ground observations with AI-processed social media intelligence into a single, real-time operational tool.

The system delivers on all primary objectives: citizens can report ocean hazards in real time via a mobile application with offline support; government officials can monitor live hazard maps with dynamic hotspot detection; analysts can extract insights from social media streams using NLP; and automated multi-level alerts ensure timely escalation to response authorities. All performance benchmarks, response times, alert latency, concurrent user support, and system uptime met or exceeded the requirements specified in the SRS.

7.2 Future Scope

Several meaningful enhancements are planned for future development phases:

- **INCOIS API Integration:** Direct integration with INCOIS early warning APIs to automatically cross-reference citizen reports with official scientific alerts, improving validation accuracy and response coordination.
- **Multilingual Support:** Extension of the NLP models to analyse social media content in regional Indian languages (Telugu, Tamil, Malayalam, Bengali) to maximize coverage in coastal communities.
- **Satellite and IoT Data Fusion:** Integration of satellite imagery feeds and IoT buoy sensor data to provide a richer, multi-source situational picture alongside crowdsourced report.
- **Federated Machine Learning:** Privacy-preserving federated learning to enable NLP model improvement using data distributed across regional agency deployments without centralizing sensitive information.

- Predictive Hotspot Modelling: Time-series forecasting models (LSTM-based) trained on historical report data to predict likely hotspot emergence during known high-risk seasonal windows.
- Gamification for Citizen Engagement: Reputation scoring, badges, and contribution leaderboards in the mobile app to incentivize consistent, accurate citizen reporting.
- AR-Based Reporting: Augmented Reality features in the mobile app enabling citizens to annotate hazard conditions directly in camera view for richer, more contextual submission.

8. REFERENCES

- Indian National Centre for Ocean Information Services (INCOIS), Ministry of Earth Sciences. Available at: <https://incois.gov.in>
- PostgreSQL Global Development Group. PostgreSQL Documentation. Available at: <https://www.postgresql.org/docs/>
- PostGIS Project. PostGIS Documentation. Available at: <https://postgis.net/documentation/>
- FastAPI. FastAPI Documentation. Available at: <https://fastapi.tiangolo.com/>
- Meta Platforms. React Documentation. Available at: <https://react.dev/>
- Google. Flutter Documentation. Available at: <https://docs.flutter.dev/>
- X (Twitter). Twitter Developer Platform. Available at: <https://developer.twitter.com/>
- Google Developers. YouTube Data API. Available at: <https://developers.google.com/youtube/v3>
- Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2018). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. arXiv:1810.04805.
- Imran, M., Castillo, C., Diaz, F., & Vieweg, S. (2015). Processing Social Media Messages in Mass Emergency: A Survey. *ACM Computing Surveys*, 47(4), 1-38.
- Goodchild, M. F., & Glennon, J. A. (2010). Crowdsourcing geographic information for disaster response: a research frontier. *International Journal of Digital Earth*, 3(3), 231-241.
- Ramsey, P. (2011). *PostGIS in Action*. Manning Publications.